Business Requirements Document

DataCo Supply Chain Analytics: Delivery Performance and Profitability

Prepared by: Manas **Tools:** Python, MySQL, Power BI **Dataset:** DataCo Smart Supply Chain for Big Data Analysis (DataCoSupplyChainDataset.csv) **Status:** Draft

1. Purpose

This document defines the business problem, objectives, scope, and deliverables for an end-to-end analytics project on DataCo Global's supply chain data. The goal is to find out where deliveries are late, where the business loses money, and what should be done about both.

2. Background

DataCo Global sells clothing, sports, and electronic supplies to customers across several markets. Its order data covers roughly 180,000 order lines and about 18,000 customers over three years, with product, customer, shipping, and financial details for each order. Management has no consolidated view of delivery performance and profitability. Late shipments hurt customer satisfaction, and a large share of orders are sold at a loss, but it is unclear where these problems are concentrated.

3. Business Problem

- A significant share of orders ship later than scheduled.
- Around 19% of order lines are loss-making.
- There is no single dashboard that shows which shipping modes, regions, products, and customer segments drive these results.

4. Objectives

1. Measure delivery performance and identify the shipping modes, regions, and markets with the highest late shipment rates.

2. Identify the product categories and departments that generate the most profit and the most losses.
3. Understand how discounts, delays, and customer segments relate to profitability.
4. Track sales, profit, and late shipment trends over time.
5. Deliver a Power BI dashboard and a written set of recommendations that a manager can act on.

5. Scope

In scope

- Cleaning and validating the raw dataset in Python.
- Feature engineering (time features, shipping delay, late shipment flag, profit margin, loss flag).
- Exploratory data analysis focused on the business questions below.
- A relational MySQL schema, data load, quality checks, analysis queries, and views.
- An interactive Power BI dashboard connected to the MySQL views.
- A project README with findings and recommendations.

Out of scope

- Real-time data pipelines or automated refresh.
- Deploying a production machine learning model.
- Analysis of the clickstream file (tokenized_access_logs.csv).
- Any use of customer personal data (masked email and password columns are dropped).

6. Stakeholders

These are the roles the analysis is written for. They are assumed for the project, not real people.

Stakeholder	Interest
Operations / Logistics Manager	Late shipments by mode, region, and market
Sales and Category Manager	Profit and loss by category and department

Stakeholder	Interest
Finance Manager	Profit margin, discount impact, loss-making orders
Senior Management	Overall KPIs and trends

7. Business Questions

Overall performance

1. What percentage of all orders are delivered late?
2. What percentage of all orders are loss-making?
3. What do sales and profit per order look like overall?

Delivery performance

4. Which shipping mode has the highest late shipment rate?
5. Which order regions have the worst late shipment rate?
6. Which markets have the worst late shipment rate?

Product profitability

7. Which categories generate the most profit?
8. Which categories lose the most money?
9. Which categories have the highest share of loss-making orders?

Customers and departments

10. Which customer segment brings in the most sales and profit?
11. Which departments are the most and least profitable?

Trends

12. How have sales and profit changed month by month and year by year?
13. Does the late shipment rate change over time?

Payments and order status

14. Which payment types are used most?

15. What share of orders are completed, canceled, or suspected fraud?

Connecting the findings

16. Do late shipments have a lower profit margin than on-time ones?
17. Do higher discounts lead to more loss-making orders?

8. Data Requirements

Item	Detail
Source	DataCo Smart Supply Chain dataset (Kaggle and Mendeley Data)
Main file	DataCoSupplyChainDataset.csv, loaded with latin-1 encoding
Data dictionary	DescriptionDataCoSupplyChain.csv
Grain	One row per order item
Key fields	Order and customer details, product and category, shipping mode, scheduled and actual shipping days, sales, discount, profit, region and market, order status, payment type

Data quality rules

- Remove columns that carry no information: masked email and password, empty product description, product image URL.
- Standardize column names to lowercase with underscores.
- Convert order and shipping dates to datetime.
- Check for duplicate order lines using the order item ID.
- Sales must be greater than zero. Negative profit is valid and is flagged, not removed.
- Shipping delay may be negative (shipped early) and is kept.

9. Functional Requirements

9.1 Python

ID	Requirement
P1	Load, inspect, and copy the dataset so the original stays untouched
P2	Clean and validate the data using the rules in section 8
P3	Create features: order year, month, day of week, shipping delay, late_shipment flag, delivery category, profit margin, is_loss flag
P4	Run EDA that answers business questions 1 to 17, with one chart per main question
P5	Check distributions, correlations, and outliers
P6	Export the cleaned dataset for MySQL

9.2 MySQL

ID	Requirement
S1	Load the cleaned CSV into a staging table
S2	Design a relational schema with customers, products, orders, order items, and shipping tables, using primary and foreign keys
S3	Run data quality checks: row counts, nulls, duplicate IDs, orphan keys
S4	Write analysis queries using joins and group-bys for the business questions
S5	Write window function queries: month-over-month growth, category rank within market, running totals, top customer per region

ID	Requirement
S6	Create views for Power BI: sales summary, delivery performance, product profit, monthly trends

9.3 Power BI

ID	Requirement
B1	Connect to the MySQL views
B2	Show KPI cards: total sales, total profit, profit margin, late shipment rate, loss-making order rate
B3	Show delivery performance by shipping mode, region, and market
B4	Show profit and loss by category and department
B5	Show monthly trends for sales, profit, and late shipment rate
B6	Add slicers for year, market, shipping mode, and customer segment
B7	Keep a consistent layout: an overview page, a delivery page, and a profitability page

10. KPIs

KPI	Definition
Late shipment rate	Share of order lines where actual shipping days exceed scheduled days
Loss-making order rate	Share of order lines with negative profit
Total sales	Sum of sales
Total profit	Sum of order profit
Profit margin	Profit divided by sales

KPI	Definition
Average shipping delay	Actual minus scheduled shipping days

11. Deliverables

1. Python notebook covering cleaning, feature engineering, and EDA.
2. Cleaned dataset (CSV).
3. MySQL scripts: schema, load, quality checks, analysis queries, window functions, and views.
4. Power BI dashboard file (.pbix) and screenshots.
5. GitHub repository with a README that states the business problem, approach, key findings, and recommendations.

12. Assumptions and Constraints

- The dataset is a published sample, so findings describe this data and are not claims about a real company.
- The late_shipment flag is based on shipping days, and may differ from the dataset's own late delivery risk column.
- The last year of data may be partial, so year-level comparisons need care.
- Delivery status and late delivery risk columns are close to the late outcome and must be excluded if any model is built.
- The project is built by one person on a local machine.

13. Success Criteria

- All 17 business questions are answered with numbers and charts.
- The MySQL schema loads cleanly with no orphan keys or duplicates.
- The Power BI dashboard runs from the MySQL views and updates correctly with the slicers.
- The README gives at least five clear findings and matching recommendations.
- A reviewer can reproduce the project from the repository.

14. Project Phases

Phase	Work
1	Python: cleaning and validation
2	Python: feature engineering
3	Python: EDA and findings
4	MySQL: schema, load, checks, queries, views
5	Power BI: dashboard
6	README and GitHub publishing